

Saketh Velidimalla

NJ, USA

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Applied AI systems engineer focused on inference engineering, evaluation harnesses, context/memory budget allocation, and trace-driven data systems for enterprise support automation. Built measurement loops that connect model behavior, retrieval quality, token usage, latency, API payloads, and downstream workflow outcomes. Interested in context-aware agent backends and cost-aware inference systems grounded in observability, regression testing, and practical system design.

Core Competencies

- **Inference & RAG Systems:** RAG pipelines, LLM workflows, agents, retrieval routing, vector search, context assembly, FastAPI
- **Evaluation & Observability:** LLM-as-judge, regression tests, Arize Phoenix, traces, hallucination/retrieval quality, RCA
- **Data & Cloud Engineering:** Python, SQL, PySpark, Airflow DAGs, DynamoDB, Databricks, Delta/Parquet, Docker, AWS Bedrock/S3/EMR/Redshift
- **Enterprise Analytics:** Power BI/DAX, semantic models, ServiceNow analytics, API debugging, metric catalogs, governance docs

Professional Experience

- **Johnson & Johnson** **New Brunswick, NJ**
Data Scientist *Feb 2025–Present*
 - Designed JAIDA routing and outcome taxonomy for right-first-time journey measurement, linking topic classification to self-service resolution, ServiceNow case creation, live-agent handoff, knowledge-base gaps, and drop-off states
 - Designed the metadata capture strategy across JAIDA middleware and cognitive-layer flows, defining which trace fields, API payload values, token usage, latency, errors, and reporting events were emitted for RCA and dashboards
 - Wrote ETL pipelines bringing JAIDA interaction data from DynamoDB into Databricks views consumed by Power BI semantic models; used schema checks, Delta/Parquet validation, and AI-assisted Databricks SQL/notebook exploration to replace ad-hoc extracts with reusable metrics tables
 - Built a Phoenix-based evaluation and observability harness using AWS Bedrock / Llama 3.2 judges, rule classifiers, keyword fallbacks, ground-truth prompts, and retrieved KB context to produce relevance, QA accuracy, hallucination, latency, error, and token-usage metrics over time
 - Connected JAIDA traces, DynamoDB records, API payloads, and ServiceNow workflow tables to model right-first-time employee journeys and separate AI behavior, routing logic, profile data, and downstream intake issues
 - Turned RCA findings into actionable KPI definitions, lineage notes, PII classification inputs, retention guidance, runbooks, operational dashboards, and leadership reporting for engineering and content ops
- **Hewlett Packard Inc.** **Spring, TX**
Data Science Engineer *Jul 2024–Nov 2024*
 - Built Airflow DAGs running PySpark transforms from S3 PC telemetry into Redshift views for Power BI reporting across hardware, software, power, sensor, and BIOS data
 - Built quarterly engineering KPI reports for hardware, software, and chip-design teams, helping stakeholders compare telemetry trends, prioritize defects, and support roadmap/funding planning
 - Optimized PySpark processing on AWS EMR, reducing runtime by about 25% through query optimization and repartitioning
- **ECrent Worldwide Company** **Remote**
Machine Learning Engineer *Jul 2021–Jun 2022*
 - Built a content-based recommendation system matching users to relevant listings with text and structured attributes; served Flask API results and evaluated ranking with Precision-at-K/Recall-at-K to improve marketplace discovery

Independent Projects

- **SakethWiki:** Local RAG and inference-evaluation harness with retrieval memory, ingestion curation, telemetry, eval replay, trace review, and model-routing experiments.
- **Buddy:** Agentic runtime prototype focused on context/memory budget allocation, context assembly, token-cost tradeoffs, retrieval routing, local LLM fallback, and unit-tested prompt behavior.

Education & Certifications

- **Arizona State University** **Phoenix, AZ**
MS Information Technology (Data Science) *Aug 2022–May 2024*
- **Completed:** Claude Code in Action, Building with Claude API, Agentic AI specialization (Andrew Ng), LLM Engineering foundations